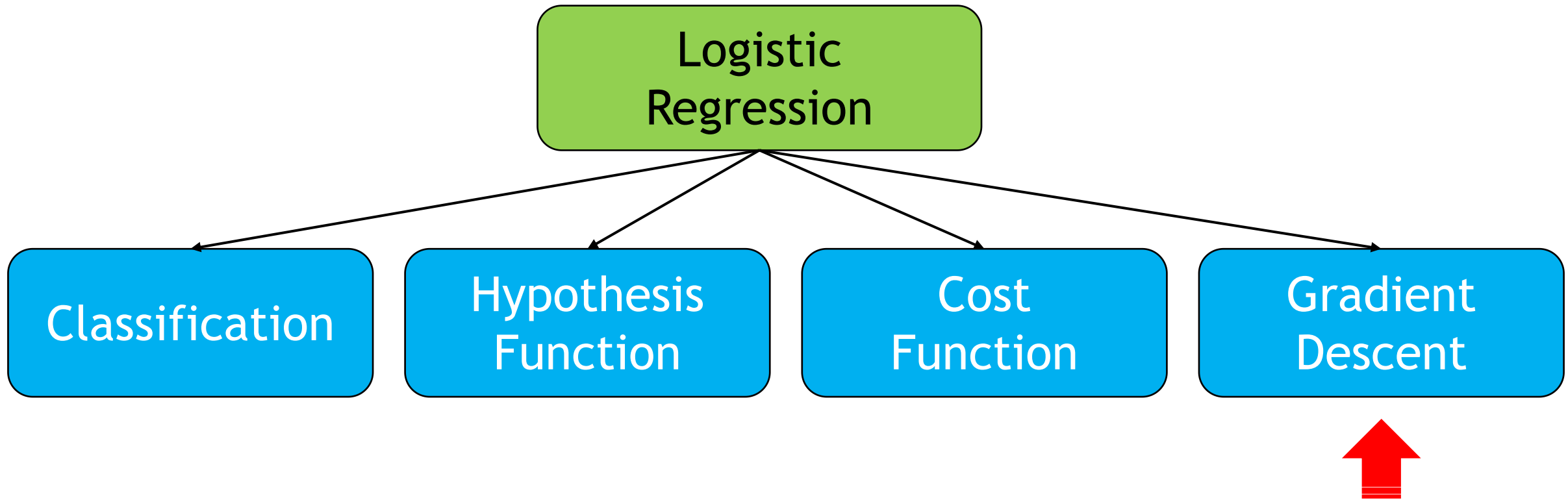


Logistic Regression- Part II

Lecture 6+7

Outline



Learning a Logistic Regression Model

- How to learn a *logistic regression model* $\mathbf{h}_{\boldsymbol{\theta}}(\mathbf{x}) = \mathbf{g}(\boldsymbol{\theta}^T \mathbf{x})$, where $\boldsymbol{\theta} = [\boldsymbol{\theta}_0, \dots, \boldsymbol{\theta}_m]$ and $\mathbf{x} = [\mathbf{x}_0, \dots, \mathbf{x}_m]$?

- By minimizing the following cost function:

$$\text{Cost}(\mathbf{h}_{\boldsymbol{\theta}}(\mathbf{x}), y) = -y \log \left(\frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}}} \right) - (1 - y) \log \left(1 - \frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}}} \right)$$

- That is:

$$\underset{\boldsymbol{\theta}}{\text{minimize}} \frac{1}{n} \sum_{i=1}^n \text{Cost}(\mathbf{h}_{\boldsymbol{\theta}}(\mathbf{x})^{(i)}, y^{(i)})$$

≡

$$\underset{\boldsymbol{\theta}}{\text{minimize}} \frac{1}{n} \sum_{i=1}^n -y^{(i)} \log \left(\frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}^{(i)}}} \right) - (1 - y^{(i)}) \log \left(1 - \frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}^{(i)}}} \right)$$

Cost function $J(\boldsymbol{\theta})$

Gradient Descent For Logistic Regression

- Outline:

- Have cost function $J(\theta)$, where $\theta = [\theta_0, \dots, \theta_m]$
- Start off with some guesses for $\theta_0, \dots, \theta_m$
 - It does not really matter what values you start off with, but a common choice is to set them all initially to zero
- Repeat until convergence{

$$\theta_j = \theta_j - \alpha \frac{\partial J(\theta)}{\partial \theta_j}$$

Partial derivative

Note: Update all θ_j simultaneously

Learning rate, which controls how big a step we take when we update θ_j

Gradient Descent For Logistic Regression

- Outline:

- Have cost function $J(\boldsymbol{\theta})$, where $\boldsymbol{\theta} = [\theta_0, \dots, \theta_m]$
- Start off with some guesses for $\theta_0, \dots, \theta_m$
 - It does not really matter what values you start off with, but a common choice is to set them all initially to zero
- Repeat until convergence{

$$\theta_j = \theta_j - \alpha \sum_{i=1}^n \left(\frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}^{(i)}}} - y^{(i)} \right) \mathbf{x}_j^{(i)}$$

The final formula after applying partial derivative.

}

Inference After Learning

- After learning the parameters $\boldsymbol{\theta} = [\theta_0, \dots, \theta_m]$, we can predict the output of any new unseen $\mathbf{x} = [\mathbf{x}_0, \dots, \mathbf{x}_m]$ as follows:

$$\left\{ \begin{array}{l} \textit{if } h_{\boldsymbol{\theta}}(\mathbf{x}) = \frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}}} < 0.5 \text{ predict } 0 \\ \textit{Else if } h_{\boldsymbol{\theta}}(\mathbf{x}) = \frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}}} \geq 0.5 \text{ predict } 1 \end{array} \right.$$

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

	and	vaccine	the	of	nigeria	y
Email a	1	1	0	1	1	1
Email b	0	0	1	1	0	0
Email c	0	1	1	0	0	1
Email d	1	0	0	1	0	0
Email e	1	0	1	0	1	1
Email f	1	0	1	1	0	0

A Training Dataset

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

	and	vaccine	the	of	nigeria	y
Email a	1	1	0	1	1	1
Email b	0	0	1	1	0	0
Email c	0	1	1	0	0	1
Email d	1	0	0	1	0	0
Email e	1	0	1	0	1	1
Email f	1	0	1	1	0	0

1 entails that a word (i.e., "and") is *present* in an email (i.e., "Email a")

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

	and	vaccine	the	of	nigeria	y
Email a	1	1	0	1	1	1
Email b	0	0	1	1	0	0
Email c	0	1	1	0	0	1
Email d	1	0	0	1	0	0
Email e	1	0	1	0	1	1
Email f	1	0	1	1	0	0

0 entails that a word (i.e., "and") is *abscent* in an email (i.e., "Email b")

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0]$
We define 6 parameters (the first one, i.e., θ_0 , is the intercept)
5 words (or *features*) = $[x_1, x_2, x_3, x_4, x_5]$

	$x_1 =$ and	$x_2 =$ vac ci ne	$x_3 =$ the	$x_4 =$ of	$x_5 =$ nigeria	y
Email a	1	1	0	1	1	1
Email b	0	0	1	1	0	0
Email c	0	1	1	0	0	1
Email d	1	0	0	1	0	0
Email e	1	0	1	0	1	1
Email f	1	0	1	1	0	0

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

$x = [x_0, x_1, x_2, x_3, x_4, x_5]$ feature vector

	$x_0 = 1$	$x_1 = \text{an d}$	$x_2 = \text{vacchi ne}$	$x_3 = \text{the}$	$x_4 = \text{of}$	$x_5 = \text{nigeria}$	y
Email a	1	1	1	0	1	1	1
Email b	1	0	0	1	1	0	0
Email c	1	0	1	1	0	0	1
Email d	1	1	0	0	1	0	0
Email e	1	1	0	1	0	1	1
Email f	1	1	0	1	1	0	0

To account for the intercept

Recap: Gradient Descent For Logistic Regression

- Outline:

- Have cost function $J(\boldsymbol{\theta})$, where $\boldsymbol{\theta} = [\theta_0, \dots, \theta_m]$
- Start off with some guesses for $\theta_0, \dots, \theta_m$
 - It does not really matter what values you start off with, but a common choice is to set them all initially to zero
- Repeat until convergence{

$$\theta_j = \theta_j - \alpha \sum_{i=1}^n \left(\frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}^{(i)}}} - y^{(i)} \right) x_j^{(i)}$$

}

First, let us calculate this factor for every example in our training dataset

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	
[1,1,1,0,1,1]	1	$[0,0,0,0,0,0] \times [1,1,1,0,1,1] = 0$	
[1,0,0,1,1,0]	0	$[0,0,0,0,0,0] \times [1,0,0,1,1,0] = 0$	
[1,0,1,1,0,0]	1	$[0,0,0,0,0,0] \times [1,0,1,1,0,0] = 0$	
[1,1,0,0,1,0]	0	$[0,0,0,0,0,0] \times [1,1,0,0,1,0] = 0$	
[1,1,0,1,0,1]	1	$[0,0,0,0,0,0] \times [1,1,0,1,0,1] = 0$	
[1,1,0,1,1,1]	0	$[0,0,0,0,0,0] \times [1,1,0,1,1,1]$	

Recap: Gradient Descent For Logistic Regression

- Outline:

- Have cost function $J(\boldsymbol{\theta})$, where $\boldsymbol{\theta} = [\theta_0, \dots, \theta_m]$
- Start off with some guesses for $\theta_0, \dots, \theta_m$
 - It does not really matter what values you start off with, but a common choice is to set them all initially to zero
- Repeat until convergence{

$$\theta_j = \theta_j - \alpha \sum_{i=1}^n \left(\frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}^{(i)}}} - y^{(i)} \right) x_j^{(i)}$$

}

Second, let us calculate this equation for every example in our training dataset and for every θ_j where j is between 0 and m

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	$(\frac{1}{1+e^{-\theta^T x}} - y) x_0$
[1,1,1,0,1,1]	1	$[0,0,0,0,0,0] \times [1,1,1,0,1,1] = 0$	
[1,0,0,1,1,0]	0	$[0,0,0,0,0,0] \times [1,0,0,1,1,0] = 0$	
[1,0,1,1,0,0]	1	$[0,0,0,0,0,0] \times [1,0,1,1,0,0] = 0$	
[1,1,0,0,1,0]	0	$[0,0,0,0,0,0] \times [1,1,0,0,1,0] = 0$	
[1,1,0,1,0,1]	1	$[0,0,0,0,0,0] \times [1,1,0,1,0,1] = 0$	
[1,1,0,1,1,1]	0	$[0,0,0,0,0,0] \times [1,1,0,1,1,1] = 0$	

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	$(\frac{1}{1+e^{-\theta^T x}} - y) x_0$
[1,1,1,0,1,1]	1	$[0,0,0,0,0,0] \times [1,1,1,0,1,1] = 0$	$(\frac{1}{1+e^{-0}} - 1) \times 1 = -0.5$
[1,0,0,1,1,0]	0	$[0,0,0,0,0,0] \times [1,0,0,1,1,0] = 0$	$(\frac{1}{1+1} - 0) \times 1 = 0.5$
[1,0,1,1,0,0]	1	$[0,0,0,0,0,0] \times [1,0,1,1,0,0] = 0$	$(\frac{1}{1+1} - 1) \times 1 = -0.5$
[1,1,0,0,1,0]	0	$[0,0,0,0,0,0] \times [1,1,0,0,1,0] = 0$	$(\frac{1}{1+1} - 0) \times 1 = 0.5$
[1,1,0,1,0,1]	1	$[0,0,0,0,0,0] \times [1,1,0,1,0,1] = 0$	$(\frac{1}{1+1} - 1) \times 1 = -0.5$
[1,1,0,1,1,1]	0	$[0,0,0,0,0,0] \times [1,1,0,1,1,1] = 0$	$(\frac{1}{1+1} - 0) \times 1 = 0.5$

Recap: Gradient Descent For Logistic Regression

- Outline:

- Have cost function $J(\boldsymbol{\theta})$, where $\boldsymbol{\theta} = [\theta_0, \dots, \theta_m]$
- Start off with some guesses for $\theta_0, \dots, \theta_m$
 - It does not really matter what values you start off with, but a common choice is to set them all initially to zero
- Repeat until convergence{

$$\theta_j = \theta_j - \alpha \sum_{i=1}^n \left(\frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}^{(i)}}} - y^{(i)} \right) x_j^{(i)}$$

Third, let us compute every θ_j

}

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	$(\frac{1}{1+e^{-\theta^T x}} - y)x_0$
[1,1,1,0,1,1]	1	$[0,0,0,0,0,0] \times [1,1,1,0,1,1] = 0$	-0.5
[1,0,0,1,1,0]	0	$[0,0,0,0,0,0] \times [1,0,0,1,1,0] = 0$	0.5
[1,0,1,1,0,0]	1	$[0,0,0,0,0,0] \times [1,0,1,1,0,0] = 0$	-0.5
[1,1,0,0,1,0]	0	$[0,0,0,0,0,0] \times [1,1,0,0,1,0] = 0$	0.5
[1,1,0,1,0,1]	1	$[0,0,0,0,0,0] \times [1,1,0,1,0,1] = 0$	-0.5
[1,1,0,1,1,1]	0	$[0,0,0,0,0,0] \times [1,1,0,1,1,1] = 0$	0.5

$$\sum_{i=1}^n \left(\frac{1}{1 + e^{-\theta^T x^{(i)}}} - y^{(i)} \right) x_0^{(i)} = 0$$

Then,

$$\theta_0 = \theta_0 - \alpha \times 0$$

New θ_0

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	$(\frac{1}{1+e^{-\theta^T x}} - y)x_0$
[1,1,1,0,1,1]	1	$[0,0,0,0,0,0] \times [1,1,1,0,1,1] = 0$	-0.5
[1,0,0,1,1,0]	0	$[0,0,0,0,0,0] \times [1,0,0,1,1,0] = 0$	0.5
[1,0,1,1,0,0]	1	$[0,0,0,0,0,0] \times [1,0,1,1,0,0] = 0$	-0.5
[1,1,0,0,1,0]	0	$[0,0,0,0,0,0] \times [1,1,0,0,1,0] = 0$	0.5
[1,1,0,1,0,1]	1	$[0,0,0,0,0,0] \times [1,1,0,1,0,1] = 0$	-0.5
[1,1,0,1,1,1]	0	$[0,0,0,0,0,0] \times [1,1,0,1,1,1] = 0$	0.5

$$\sum_{i=1}^n \left(\frac{1}{1 + e^{-\theta^T x^{(i)}}} - y^{(i)} \right) x_0^{(i)} = 0$$

Then,

$$\theta_0 = \theta_0 - \alpha \times 0$$

Old θ_0

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	$(\frac{1}{1+e^{-\theta^T x}} - y) x_0$
[1,1,1,0,1,1]	1	$[0,0,0,0,0,0] \times [1,1,1,0,1,1] = 0$	-0.5
[1,0,0,1,1,0]	0	$[0,0,0,0,0,0] \times [1,0,0,1,1,0] = 0$	0.5
[1,0,1,1,0,0]	1	$[0,0,0,0,0,0] \times [1,0,1,1,0,0] = 0$	-0.5
[1,1,0,0,1,0]	0	$[0,0,0,0,0,0] \times [1,1,0,0,1,0] = 0$	0.5
[1,1,0,1,0,1]	1	$[0,0,0,0,0,0] \times [1,1,0,1,0,1] = 0$	-0.5
[1,1,0,1,1,1]	0	$[0,0,0,0,0,0] \times [1,1,0,1,1,1] = 0$	0.5

$$\sum_{i=1}^n \left(\frac{1}{1 + e^{-\theta^T x^{(i)}}} - y^{(i)} \right) x_0^{(i)} = 0$$

Then,

$$\begin{aligned} \theta_0 &= \theta_0 - \alpha \times 0 \\ &= 0 - 0.5 \times 0 = 0 \end{aligned}$$

New Parameter Vector:

$$\theta = [0, \theta_1, \theta_2, \theta_3, \theta_4, \theta_5]$$

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	$(\frac{1}{1+e^{-\theta^T x}} - y)x_1$
[1,1,1,0,1,1]	1	$[0,0,0,0,0,0] \times [1,1,1,0,1,1] = 0$	
[1,0,0,1,1,0]	0	$[0,0,0,0,0,0] \times [1,0,0,1,1,0] = 0$	
[1,0,1,1,0,0]	1	$[0,0,0,0,0,0] \times [1,0,1,1,0,0] = 0$	
[1,1,0,0,1,0]	0	$[0,0,0,0,0,0] \times [1,1,0,0,1,0] = 0$	
[1,1,0,1,0,1]	1	$[0,0,0,0,0,0] \times [1,1,0,1,0,1] = 0$	
[1,1,0,1,1,1]	0	$[0,0,0,0,0,0] \times [1,1,0,1,1,1] = 0$	

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	$(\frac{1}{1+e^{-\theta^T x}} - y)x_1$
[1,1,1,0,1,1]	1	$[0,0,0,0,0,0] \times [1,1,1,0,1,1] = 0$	-0.5
[1,0,0,1,1,0]	0	$[0,0,0,0,0,0] \times [1,0,0,1,1,0] = 0$	0
[1,0,1,1,0,0]	1	$[0,0,0,0,0,0] \times [1,0,1,1,0,0] = 0$	0
[1,1,0,0,1,0]	0	$[0,0,0,0,0,0] \times [1,1,0,0,1,0] = 0$	0.5
[1,1,0,1,0,1]	1	$[0,0,0,0,0,0] \times [1,1,0,1,0,1] = 0$	-0.5
[1,1,0,1,1,1]	0	$[0,0,0,0,0,0] \times [1,1,0,1,1,1] = 0$	-0.5

$$\sum_{i=1}^n \left(\frac{1}{1 + e^{-\theta^T x^{(i)}}} - y^{(i)} \right) x_1^{(i)} = 0$$

Then,

$$\begin{aligned} \theta_1 &= \theta_1 - \alpha \times 0 \\ &= 0 - 0.5 \times 0 = 0 \end{aligned}$$

New Parameter Vector:
 $\theta = [0, 0, \theta_2, \theta_3, \theta_4, \theta_5]$

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	$(\frac{1}{1+e^{-\theta^T x}} - y)x_2$
[1,1,1,0,1,1]	1	$[0,0,0,0,0,0] \times [1,1,1,0,1,1] = 0$	
[1,0,0,1,1,0]	0	$[0,0,0,0,0,0] \times [1,0,0,1,1,0] = 0$	
[1,0,1,1,0,0]	1	$[0,0,0,0,0,0] \times [1,0,1,1,0,0] = 0$	
[1,1,0,0,1,0]	0	$[0,0,0,0,0,0] \times [1,1,0,0,1,0] = 0$	
[1,1,0,1,0,1]	1	$[0,0,0,0,0,0] \times [1,1,0,1,0,1] = 0$	
[1,1,0,1,1,1]	0	$[0,0,0,0,0,0] \times [1,1,0,1,1,1] = 0$	

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	$(\frac{1}{1+e^{-\theta^T x}} - y)x_2$
[1,1,1,0,1,1]	1	$[0,0,0,0,0,0] \times [1,1,1,0,1,1] = 0$	-0.5
[1,0,0,1,1,0]	0	$[0,0,0,0,0,0] \times [1,0,0,1,1,0] = 0$	0
[1,0,1,1,0,0]	1	$[0,0,0,0,0,0] \times [1,0,1,1,0,0] = 0$	-0.5
[1,1,0,0,1,0]	0	$[0,0,0,0,0,0] \times [1,1,0,0,1,0] = 0$	0
[1,1,0,1,0,1]	1	$[0,0,0,0,0,0] \times [1,1,0,1,0,1] = 0$	0
[1,1,0,1,1,1]	0	$[0,0,0,0,0,0] \times [1,1,0,1,1,1] = 0$	0

$$\sum_{i=1}^n \left(\frac{1}{1 + e^{-\theta^T x^{(i)}}} - y^{(i)} \right) x_2^{(i)} = -1$$

Then,

$$\theta_2 = \theta_2 - \alpha \times (-1)$$

$$= 0 - 0.5 \times (-1) = 0.5$$

New Parameter Vector:

$$\theta = [0, 0, 0.5, \theta_3, \theta_4, \theta_5]$$

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	$(\frac{1}{1+e^{-\theta^T x}} - y)x_3$
[1,1,1,0,1,1]	1	$[0,0,0,0,0,0] \times [1,1,1,0,1,1] = 0$	
[1,0,0,1,1,0]	0	$[0,0,0,0,0,0] \times [1,0,0,1,1,0] = 0$	
[1,0,1,1,0,0]	1	$[0,0,0,0,0,0] \times [1,0,1,1,0,0] = 0$	
[1,1,0,0,1,0]	0	$[0,0,0,0,0,0] \times [1,1,0,0,1,0] = 0$	
[1,1,0,1,0,1]	1	$[0,0,0,0,0,0] \times [1,1,0,1,0,1] = 0$	
[1,1,0,1,1,1]	0	$[0,0,0,0,0,0] \times [1,1,0,1,1,1]$	

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	$(\frac{1}{1+e^{-\theta^T x}} - y) x_3$
[1,1,1,0,1,1]	1	[0,0,0,0,0,0]×[1,1,1,0,1,1]=0	0
[1,0,0,1,1,0]	0	[0,0,0,0,0,0]×[1,0,0,1,1,0]=0	0.5
[1,0,1,1,0,0]	1	[0,0,0,0,0,0]×[1,0,1,1,0,0]=0	-0.5
[1,1,0,0,1,0]	0	[0,0,0,0,0,0]×[1,1,0,0,1,0]=0	0
[1,1,0,1,0,1]	1	[0,0,0,0,0,0]×[1,1,0,1,0,1]=0	-0.5
[1,1,0,1,1,1]	0	[0,0,0,0,0,0]×[1,1,0,1,1,1]	-0.5

$$\sum_{i=1}^n \left(\frac{1}{1 + e^{-\theta^T x^{(i)}}} - y^{(i)} \right) x_3^{(i)} = 0$$

Then,

$$\begin{aligned} \theta_3 &= \theta_3 - \alpha \times 0 \\ &= 0 - 0.5 \times 0 = 0 \end{aligned}$$

New Parameter Vector:
 $\theta = [0, 0, 0.5, 0, \theta_4, \theta_5]$

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	$(\frac{1}{1+e^{-\theta^T x}} - y)x_4$
$[1, 1, 1, 0, 1, 1]$	1	$[0, 0, 0, 0, 0, 0] \times [1, 1, 1, 0, 1, 1] = 0$	
$[1, 0, 0, 1, 1, 0]$	0	$[0, 0, 0, 0, 0, 0] \times [1, 0, 0, 1, 1, 0] = 0$	
$[1, 0, 1, 1, 0, 0]$	1	$[0, 0, 0, 0, 0, 0] \times [1, 0, 1, 1, 0, 0] = 0$	
$[1, 1, 0, 0, 1, 0]$	0	$[0, 0, 0, 0, 0, 0] \times [1, 1, 0, 0, 1, 0] = 0$	
$[1, 1, 0, 1, 0, 1]$	1	$[0, 0, 0, 0, 0, 0] \times [1, 1, 0, 1, 0, 1] = 0$	
$[1, 1, 0, 1, 1, 1]$	0	$[0, 0, 0, 0, 0, 0] \times [1, 1, 0, 1, 1, 1] = 0$	

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	$(\frac{1}{1+e^{-\theta^T x}} - y) x_4$
[1,1,1,0,1,1]	1	$[0,0,0,0,0,0] \times [1,1,1,0,1,1] = 0$	-0.5
[1,0,0,1,1,0]	0	$[0,0,0,0,0,0] \times [1,0,0,1,1,0] = 0$	0.5
[1,0,1,1,0,0]	1	$[0,0,0,0,0,0] \times [1,0,1,1,0,0] = 0$	0
[1,1,0,0,1,0]	0	$[0,0,0,0,0,0] \times [1,1,0,0,1,0] = 0$	0.5
[1,1,0,1,0,1]	1	$[0,0,0,0,0,0] \times [1,1,0,1,0,1] = 0$	0
[1,1,0,1,1,1]	0	$[0,0,0,0,0,0] \times [1,1,0,1,1,1] = 0$	0.5

$$\sum_{i=1}^n \left(\frac{1}{1 + e^{-\theta^T x^{(i)}}} - y^{(i)} \right) x_4^{(i)} = 1$$

Then,

$$\begin{aligned} \theta_4 &= \theta_4 - \alpha \times 1 \\ &= 0 - 0.5 \times 1 = -0.5 \end{aligned}$$

New Parameter Vector:

$$\theta = [0, 0, 0.5, 0, -0.5, \theta_5]$$

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	$(\frac{1}{1+e^{-\theta^T x}} - y) x_5$
[1,1,1,0,1,1]	1	[0,0,0,0,0,0]×[1,1,1,0,1,1]=0	
[1,0,0,1,1,0]	0	[0,0,0,0,0,0]×[1,0,0,1,1,0]=0	
[1,0,1,1,0,0]	1	[0,0,0,0,0,0]×[1,0,1,1,0,0]=0	
[1,1,0,0,1,0]	0	[0,0,0,0,0,0]×[1,1,0,0,1,0]=0	
[1,1,0,1,0,1]	1	[0,0,0,0,0,0]×[1,1,0,1,0,1]=0	
[1,1,0,1,1,1]	0	[0,0,0,0,0,0]×[1,1,0,1,1,1]	

A Concrete Example: The Training Phase

- Let us apply logistic regression on the spam email recognition problem, assuming $\alpha = 0.5$ and starting with $\theta = [0, 0, 0, 0, 0, 0]$

x	y	$\theta^T x$	$\left(\frac{1}{1+e^{-\theta^T x}} - y\right) x_5$
[1,1,1,0,1,1]	1	$[0,0,0,0,0,0] \times [1,1,1,0,1,1] = 0$	-0.5
[1,0,0,1,1,0]	0	$[0,0,0,0,0,0] \times [1,0,0,1,1,0] = 0$	0
[1,0,1,1,0,0]	1	$[0,0,0,0,0,0] \times [1,0,1,1,0,0] = 0$	0
[1,1,0,0,1,0]	0	$[0,0,0,0,0,0] \times [1,1,0,0,1,0] = 0$	0
[1,1,0,1,0,1]	1	$[0,0,0,0,0,0] \times [1,1,0,1,0,1] = 0$	-0.5
[1,1,0,1,1,1]	0	$[0,0,0,0,0,0] \times [1,1,0,1,1,1]$	

$$\sum_{i=1}^n \left(\frac{1}{1 + e^{-\theta^T x^{(i)}}} - y^{(i)} \right) x_5^{(i)} = -1$$

Then,

$$\theta_5 = \theta_5 - \alpha \times (-1)$$

$$= 0 - 0.5 \times (-1) = 0.5$$

New Parameter Vector:

$$\theta = [0, 0, 0.5, 0, -0.5, 0.5]$$

A Concrete Example: Testing

- Let us now *test* logistic regression on the spam email recognition problem, using the just learnt $\theta = [0, 0, 0.5, 0, -0.5, 0.5]$
 - **Note:** Testing is typically done over a portion of the dataset that is not used during training, but rather kept only for testing the accuracy of the algorithm's predictions thus far
 - In this example, we will test over all the examples that we *used* during training, *just* for illustrative purposes

A Concrete Example: Testing

- Let us *test* logistic regression on the spam email recognition problem, using the just learnt $\theta = [0, 0, 0.5, 0, -0.5, 0.5]$

x	y	$\theta^T x$
$[1, 1, 1, 0, 1, 1]$	1	$[0, 0, 0.5, 0, -0.5, 0.5] \times [1, 1, 1, 0, 1, 1] = 0.5$
$[1, 0, 0, 1, 1, 0]$	0	$[0, 0, 0.5, 0, -0.5, 0.5] \times [1, 0, 0, 1, 1, 0] = -0.5$
$[1, 0, 1, 1, 0, 0]$	1	$[0, 0, 0.5, 0, -0.5, 0.5] \times [1, 0, 1, 1, 0, 0] = 0.5$
$[1, 1, 0, 0, 1, 0]$	0	$[0, 0, 0.5, 0, -0.5, 0.5] \times [1, 1, 0, 0, 1, 0] = -0.5$
$[1, 1, 0, 1, 0, 1]$	1	$[0, 0, 0.5, 0, -0.5, 0.5] \times [1, 1, 0, 1, 0, 1] = 0.5$

A Concrete Example: Testing

- Let us *test* logistic regression on the spam email recognition problem, using the just learnt $\theta = [0, 0, 0.5, 0, -0.5, 0.5]$

x	y	$\theta^T x$	$h_{\theta}(x) = \left(\frac{1}{1+e^{-\theta^T x}} \right)$
[1,1,1,0,1,1]	1	$[0,0,0.5,0,-0.5,0.5] \times [1,1,1,0,1,1] = 0.5$	0.622459331
[1,0,0,1,1,0]	0	$[0,0,0.5,0,-0.5,0.5] \times [1,0,0,1,1,0] = -0.5$	0.377540669
[1,0,1,1,0,0]	1	$[0,0,0.5,0,-0.5,0.5] \times [1,0,1,1,0,0] = 0.5$	0.622459331
[1,1,0,0,1,0]	0	$[0,0,0.5,0,-0.5,0.5] \times [1,1,0,0,1,0] = -0.5$	0.377540669
[1,1,0,1,0,1]	1	$[0,0,0.5,0,-0.5,0.5] \times [1,1,0,1,0,1] = 0.5$	0.622459331

A Concrete Example: Testing

- Let us *test* logistic regression on the spam email recognition problem, using the just learnt $\theta = [0, 0, 0.5, 0, -0.5, 0.5]$
(if $h_{\theta}(x) \geq 0.5$, $y' = 1$; else y')

x	y	$\theta^T x$	$h_{\theta}(x) = \left(\frac{1}{1+e^{-\theta^T x}} \right)$	Predicted Class (or y')
[1,1,1,0,1,1]	1	$[0,0,0.5,0,-0.5,0.5] \times [1,1,1,0,1,1] = 0.5$	0.622459331	
[1,0,0,1,1,0]	0	$[0,0,0.5,0,-0.5,0.5] \times [1,0,0,1,1,0] = -0.5$	0.377540669	
[1,0,1,1,0,0]	1	$[0,0,0.5,0,-0.5,0.5] \times [1,0,1,1,0,0] = 0.5$	0.622459331	
[1,1,0,0,1,0]	0	$[0,0,0.5,0,-0.5,0.5] \times [1,1,0,0,1,0] = -0.5$	0.377540669	
[1,1,0,1,0,1]	1	$[0,0,0.5,0,-0.5,0.5] \times [1,1,0,1,0,1] = 0.5$	0.622459331	

A Concrete Example: Testing

- Let us *test* logistic regression on the spam email recognition problem, using the just learnt $\theta = [0, 0, 0.5, 0, -0.5, 0.5]$
(if $h_{\theta}(x) \geq 0.5$, $y' = 1$; else y')

x	y	$\theta^T x$	$h_{\theta}(x) = \left(\frac{1}{1+e^{-\theta^T x}} \right)$	Predicted Class (or y')
[1,1,1,0,1,1]	1	$[0,0,0.5,0,-0.5,0.5] \times [1,1,1,0,1,1] = 0.5$	0.622459331	1
[1,0,0,1,1,0]	0	$[0,0,0.5,0,-0.5,0.5] \times [1,0,0,1,1,0] = -0.5$	0.377540669	0
[1,0,1,1,0,0]	1	$[0,0,0.5,0,-0.5,0.5] \times [1,0,1,1,0,0] = 0.5$	0.622459331	1
[1,1,0,0,1,0]	0	$[0,0,0.5,0,-0.5,0.5] \times [1,1,0,0,1,0] = -0.5$	0.377540669	0
[1,1,0,1,0,1]	1	$[0,0,0.5,0,-0.5,0.5] \times [1,1,0,1,0,1] = 0.5$	0.622459331	1

A Concrete Example: Testing

- Let us *test* logistic regression on the spam email recognition problem, using the just learnt $\theta = [0, 0, 0.5, 0, -0.5, 0.5]$
(if $h_{\theta}(x) \geq 0.5$, $y' = 1$; else y')

x	y	$\theta^T x$	$h_{\theta}(x) = \left(\frac{1}{1+e^{-\theta^T x}} \right)$	Predicted Class (or y')
[1,1,1,0,1,1]	1	$[0,0,0.5,0,-0.5,0.5] \times [1,1,1,0,1,1] = 0.5$	0.622459331	1
[1,0,0,1,1,0]	0	$[0,0,0.5,0,-0.5,0.5] \times [1,0,0,1,1,0] = -0.5$	0.377540669	NO
[1,0,1,1,0,0]	1	$[0,0,0.5,0,-0.5,0.5] \times [1,0,1,1,0,0] = 0.5$	0.622459331	1
[1,1,0,0,1,0]	0	$[0,0,0.5,0,-0.5,0.5] \times [1,1,0,0,1,0] = -0.5$	0.377540669	0
[1,1,0,1,0,1]	1	$[0,0,0.5,0,-0.5,0.5] \times [1,1,0,1,0,1] = 0.5$	0.622459331	1

Mispredictions!

A Concrete Example: Inference

- Let us infer whether a given new email, say, $k = [1, 0, 1, 0, 0, 1]$ is a spam or not, using logistic regression with the just learnt parameter vector

$$\theta = [0 \quad 0 \quad 0 \quad 5 \quad 0 \quad -0.5 \quad 0.5]$$

	$x_0 = 1$	$x_1 = \text{an d}$	$x_2 = \text{vacchi ne}$	$x_3 = \text{the}$	$x_4 = \text{of}$	$x_5 = \text{nigeria}$	y
Email a	1	1	1	0	1	1	1
Email b	1	0	0	1	1	0	0
Email c	1	0	1	1	0	0	1
Email d	1	1	0	0	1	0	0
Email e	1	1	0	1	0	1	1

Our Training Dataset

A Concrete Example: Inference

- Let us infer whether a given new email, say, $k = [1, 0, 1, 0, 0, 1]$ is a spam or not, using logistic regression with the just learnt parameter vector

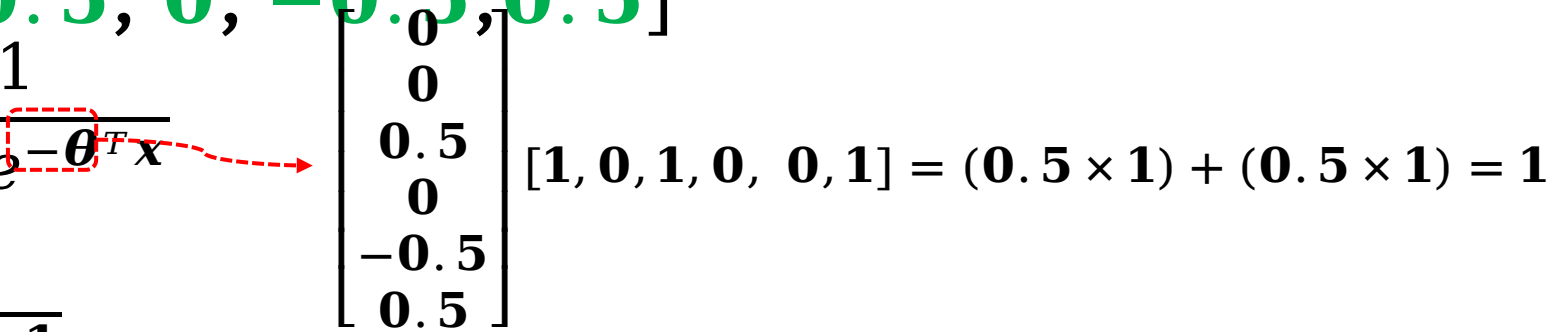
$$\theta = [0 \quad 0 \quad 0 \quad 5 \quad 0 \quad -0.5 \quad 0.5]$$

	$x_0 = 1$	$x_1 = \text{an}$ d	$x_2 = \text{vacci}$ ne	$x_3 = \text{the}$	$x_4 =$ of	$x_5 = \text{nigeria}$	y
Email a	1	1	1	0	1	1	1
Email b	1	0	0	1	1	0	0
Email c	1	0	1	1	0	0	1
Email d	1	1	0	0	1	0	0
Email e	1	1	0	1	0	1	1
Email f	1	1	0	1	1	0	0
Email k	1	0	1	0	0	1	?

A Concrete Example: Inference

- Let us infer whether a given new email, say, $k = [1, 0, 1, 0, 0, 1]$ is a spam or not, using logistic regression with the just learnt parameter vector

$$\theta = [0, 0, 0.5, 0, -0.5, 0.5]$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

$$= \frac{1}{1 + e^{-1}}$$

$$= 0.731$$

$$\geq 0.5 \quad \text{Class 1 (i.e., Spam)}$$

A Concrete Example: Inference

- Let us infer whether a given new email, say, $k = [1, 0, 1, 0, 0, 1]$ is a spam or not, using logistic regression with the just learnt parameter vector

$$\theta = [0 \quad 0 \quad 0 \quad 5 \quad 0 \quad -0.5 \quad 0 \quad 5]$$

	$x_0 = 1$	$x_1 = \text{an}$ d	$x_2 = \text{vacci}$ ne	$x_3 = \text{the}$	$x_4 =$ of	$x_5 = \text{nigeria}$	y
Email a	1	1	1	0	1	1	1
Email b	1	0	0	1	1	0	0
Email c	1	0	1	1	0	0	1
Email d	1	1	0	0	1	0	0
Email e	1	1	0	1	0	1	1
Email f	1	1	0	1	1	0	0
Email k	1	0	1	0	0	1	1

Somehow interesting since it considered “vaccine” and “nigeria” indicative of spam!